

# Arnab Mandal

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## EDUCATION

### Shiv Nadar University

B.Tech in Computer Science Engineering (8.59/10.0)

Noida, Delhi-NCR, India

Aug 2023 - Jul 2027

### Heera Lal Public School, Class XII

Percentage : 91%

Madan Pur Dabas, Delhi-NCR, India

May 2023

### Navy Children School, Class X

Percentage : 97%

Visakhapatnam, Andhra Pradesh, India

May 2021

## INTERNSHIPS & EXPERIENCE

### AI Engineer Intern, DeepLure Research Pvt Ltd, Amravati

Dec 2025 - Jan 2026

- Optimized Llava 1.5 Pipeline serving **50+** clients by migrating inference to TensorRT alongside Triton inference server.
- Business Impact:** Achieved GPU utilization increase by **30%** and serving **8x** clients with no additional resources.

### AI Intern, Critical AI Pvt Ltd, Bengaluru

Jul 2025 - Sep 2025

- Deployed multimodal RAG pipelines (FAISS, llama-cpp, OSM, TTS, STT, MongoDB) behind FastAPI microservices.
- Business Impact:** Added **3+** features and transitioned to on-prem LLM stack, reducing inference cost by **30%**.

### AI Intern, WESEE, Ministry of Defence, New Delhi

May 2025 - Jul 2025

- Integrated MCP agents for SQL-based analytics and fine-tuned YOLOv8 pipelines (**94%** mAP@50, **68%** mAP50-95).
- Business Impact:** Cut SME report prep time by **~90%** leveraging **prompt engineering**.

## PROJECTS

### Graph-RAG Text-to-SQL Agent for Enterprise Databases 🧠

- Engineered Text-to-SQL agent using LangGraph, tested over **43** tables and **62** relationships in a **1GB** warehouse.
- Boosted multi-join query Execution Accuracy by **33%** by orchestrating a graph-supervised, RAG-indexed pipeline.
- Technology Stack: LangChain, LangGraph, Gemini, NetworkX, React.js

### AutoLinks: Intelligently InLinking Draft Articles for SEO 🧠🔗

- Developed a semantic internal linking API to optimize site-wide link equity distribution via NER and vector search.
- Utilized LLM-as-a-judge, achieving **85+%** acceptance, achieving **p95** Latency of **2.1s**, lowering Gini coefficient by **20%**.
- Technology Stack: FastAPI, GLiNER, sentence-transformers, trafilatura, Qdrant, React.js

### Multilingual Multimodal Sentiment Analysis Engine 🧠

- Modeled a FastAPI-based engine supporting 5 languages, achieving a **0.71 F1-score** on imbalanced multimodal datasets.
- Optimized inference pipeline to handle audio+text inputs, maintaining **sub-250ms** latency for real-time analysis.
- Technology Stack: PyTorch, scikit-learn, FastAPI, transformer

### End-to-End LLM Pretraining and Adaptation pipeline 🧠🤖

- Implemented GPT-2 (MHA, GELU) from scratch; KV-cache raised throughput **5x** on RTX 4060.
- Fine-tuned LLaMA 3 8B using LoRA (PEFT) and quantized to Q5K-M GGUF, reducing memory footprint by **60-65%**.
- Technology Stack: PyTorch, TikToken, Unsloth, Transformers, PEFT, HuggingFace

## TECHNICAL SKILLS

### Languages

Python, SQL, Rust, TypeScript

### AI & ML

PyTorch, Transformers, LangChain, LangGraph, LlamaIndex, CrewAI, Optuna, vLLM

### DevOps & MLOps

FastAPI, Kubernetes, Jenkins, Docker, MLflow, DVC, Comet-ML, Triton, TensorRT, LangSmith

### Cloud

GCP, AWS, Azure

## ACHIEVEMENTS

- Smart SNU Hackathon'25 Winner:** Led a 6-member team to build and deploy the Krishi AI Sahayak app.
- LeetCode Knight:** Achieved a **1,894 peak contest rating** and solved **300+ DSA problems**.
- Dean's list: Monsoon'23**
- Semifinalist speaker at SMVIT Parliamentary Debate'24**